

CONCLUSIONS FROM AN ATTEMPT AT THE LAX–PHILLIPS–PAVLOV–FADDEEV REGULARIZED ESTIMATE

OPENAI CODEX

ABSTRACT. We report the outcome of an attempt to prove the regularized Lax–Phillips–Pavlov–Faddeev estimate associated with the modular automorphic wave equation. The physical identifications can be made precise: the final double-primed interaction semigroup is boundedly similar to an adjoint compressed-translation model for the pure Blaschke inner function $\Theta_0(s) = \xi(2s)/\xi(-2s)$, together with a one-dimensional eigenmode at $-1/2$. We obtain an exact inverse-Laplace tail formula, an H^∞ quotient formula, a confluent Nevanlinna–Pick criterion retaining all zero multiplicities, and equivalent completely positive and Lyapunov formulations. For the actual modular semigroup we prove the unconditional qualitative decay

$$\|Z(t)(B + iI)^{-1}\| \longrightarrow 0, \quad -\frac{1}{4} \leq \omega_{\text{reg}}(B) \leq 0.$$

We also construct pure meromorphic Blaschke models whose entire divisor lies on $\Re s = 1/4$ but whose regularized exponent is 0. Hence critical spectral location, innerness, meromorphicity, divisor symmetry, and root completeness do not by themselves yield the desired exponent. Complete positivity does not remove the obstruction: at the required scale it is exactly equivalent to uniform positivity of all confluent Pick matrices. The remaining problem is a genuinely uniform, zeta-specific root-synthesis or weighted-observability estimate. No unconditional proof or disproof of the target estimate is obtained, and no claim concerning a proof of the Riemann hypothesis is made.

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1. OBJECTIVE, VERDICT, AND STATUS OF POSSIBLE NOVELTY

After fixing the positive energy space, the final double-primed Lax–Phillips interaction semigroup $Z(t)$, and its ordinary C_0 -generator B , the exact objective was to prove unconditionally

$$(1) \quad \forall \varepsilon > 0 \exists C_\varepsilon < \infty \forall t \geq 0 : \quad \left\| Z(t)(B + iI)^{-1} \right\| \leq C_\varepsilon e^{(-1/4+\varepsilon)t}.$$

The norm is the operator norm from the positive energy space to itself. It is equivalently the norm of $Z(t)$ from $\text{Dom } B$, equipped with $f \mapsto \|(B + iI)f\|$, into the energy space. Neither the unregularized norm nor a fixed-vector estimate is used below.

The estimate (1) was not proved or disproved. The identification hypotheses in the governing specification were, however, verified against the native constructions, and several exact reductions and obstructions were obtained. The strongest conclusions are ordered by their mathematical significance rather than by the order in which they were found.

Remark 1.1 (Novelty convention). Several elementary consequences below were not located in the sources inspected during this project. We call these *candidate observations* and highlight them explicitly, but make no priority or originality claim. The main candidates are Theorem 2.1, Theorem 3.1, Proposition 6.1, and Theorem 9.1, together with their zeta-model synthesis. A dedicated literature search would be required before any novelty claim.

2. A CRITICAL-LINE MODEL CAN HAVE REGULARIZED EXPONENT ZERO

The following abstract example is the sharpest obstruction found. It shows that even placing every model eigenvalue on the critical spectral line does not control the regularized exponent without quantitative information on root blocks and their synthesis.

Let Θ be a pure inner function on $\mathbb{C}_+ = \{s : \text{Re } s > 0\}$ and put

$$K_\Theta = H^2(\mathbb{C}_+) \ominus \Theta H^2(\mathbb{C}_+), \quad S_\Theta(t) = \left(P_{K_\Theta} M_{e^{-ts}}|_{K_\Theta} \right)^*.$$

Write B_Θ for the ordinary generator of $S_\Theta(t)$.

Theorem 2.1 (Critical-line Blaschke counterexample). *There exists a pure meromorphic Blaschke product Θ such that:*

- (i) *every zero of Θ lies on $\text{Re } a = 1/4$, and the divisor is invariant under conjugation and under $a \mapsto 1/2 - \bar{a}$;*
- (ii) *Θ has zero mean type and no finite boundary singularity;*
- (iii) *$S_\Theta(t)$ is a contraction semigroup, $-i \in \rho(B_\Theta)$, and the generalized root kernels are complete;*
- (iv) *nevertheless,*

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log \left\| S_\Theta(t)(B_\Theta + iI)^{-1} \right\| = 0.$$

Thus critical-line spectral location, pure innerness, meromorphicity, and the centered divisor symmetries do not imply the exponent $-1/4$.

Proof. Put $\delta = 1/4$, $k_n = n$ for $n \geq 2$, and $Y_n = e^{\sqrt{n}}$. Let Θ be the pure half-plane Blaschke product whose zeros are

$$a_n^\pm = \delta \pm iY_n,$$

each with multiplicity $k_n + 1$. The Blaschke sum converges because

$$\sum_n \frac{2(k_n + 1)\delta}{1 + \delta^2 + Y_n^2} < \infty.$$

There is no exponential or finite-boundary singular factor by construction, and there is no finite boundary accumulation. The divisor has both stated symmetries.

The zero-kernel derivatives are complete in K_Θ : if $F \in K_\Theta$ is orthogonal to every derivative kernel, then F vanishes on the full zero divisor, so $F/\Theta \in H^2(\mathbb{C}_+)$. Hence $F \in K_\Theta \cap \Theta H^2 = \{0\}$. The standard model-spectrum theorem [9] says that its finite generator spectral points are the zero-derived eigenvalues together with finite boundary singularities. There are no boundary singularities here and no zero corresponds to $-i$; therefore $-i \in \rho(B_\Theta)$.

Under the unitary Laplace transform, $S_\Theta(t)$ becomes left translation on a closed subspace $\mathcal{K}_\Theta \subset L^2(0, \infty)$ and $B_\Theta = d/dx$. Since a_n^+ has multiplicity $k_n + 1$, the normalized root vector

$$f_n(x) = c_n x^{k_n} e^{-\delta x} e^{iY_n x}, \quad c_n^2 = \frac{(2\delta)^{2k_n+1}}{(2k_n)!},$$

belongs to $\mathcal{K}_\Theta \cap W^{1,2}(0, \infty)$ and has norm one. Direct calculation gives

$$(2) \quad \|(B_\Theta + iI)f_n\|_2^2 = (Y_n + 1)^2 + \frac{\delta^2}{2k_n - 1}.$$

The probability density $|f_n(x)|^2 dx$ is gamma with shape $2k_n + 1$ and rate 2δ . At

$$t_n = \frac{2k_n + 1}{4\delta},$$

Chebyshev's inequality yields

$$(3) \quad \int_{t_n}^\infty |f_n(x)|^2 dx \geq 1 - \frac{4}{2k_n + 1}.$$

Equations (2)–(3) imply

$$\|S_\Theta(t_n)(B_\Theta + iI)^{-1}\| \geq \frac{\sqrt{1 - 4/(2k_n + 1)}}{\sqrt{(Y_n + 1)^2 + \delta^2/(2k_n - 1)}}.$$

Since $\log Y_n = \sqrt{n} = o(t_n)$, the logarithmic limsup is at least zero. Contractivity and boundedness of the fixed resolvent give the reverse inequality.

The meromorphic model has no finite boundary spectrum. None of the stated qualitative properties changes the calculation above. $\square$

Remark 2.2 (Scope of the counterexample). This is not the zeta inner function and is not a counterexample to RH or to the Pavlov–Faddeev theorem. Its purpose is logical: no argument using only innerness, meromorphic continuation, functional-equation-type divisor symmetry, completeness, and critical spectral location can prove (1). The zeta case must use arithmetic information not present in the abstract model axioms.

3. THE STRONGEST UNCONDITIONAL RESULT FOR THE MODULAR SEMIGROUP

The exact Uetake model and its compact-resolvent theorem yield a genuine operator-norm conclusion, although without a negative exponential rate.

Theorem 3.1 (Qualitative regularized decay). *For the final double-primed modular interaction semigroup and its ordinary generator,*

$$(4) \quad \|Z(t)(B + iI)^{-1}\| \longrightarrow 0 \quad (t \rightarrow \infty).$$

Moreover $t \mapsto \|Z(t)(B + iI)^{-1}\|$ is nonincreasing, and

$$(5) \quad -\frac{1}{4} \leq \omega_{\text{reg}}(B) \leq 0.$$

Proof. Uetake proves that the reduced generator has meromorphic resolvent, that its generalized root vectors have dense finite linear span, and that every finite spectral point is a zeta-derived root eigenvalue; see [13, Thms. 4.2 and 4.4]. Every such eigenvalue has strictly negative real part. On a root vector the semigroup orbit is a polynomial in t times $e^{\lambda t}$, hence tends to zero. Contractivity extends strong stability from the dense root span to the whole reduced model. The full model adds only the mode $e^{-t/2}$, so the physical semigroup is strongly stable as well.

The generator resolvent is compact by the Lax–Phillips compact-resolvent theorem, reported in [13, p. 100]; see also [6, Thm. 6.17]. Uniformly bounded strong convergence is uniform on relatively compact sets. Applying it to the image of the unit ball under $(B + iI)^{-1}$ proves (4).

Since the resolvent commutes with $Z(t)$ and $Z(t)$ is contractive,

$$\|Z(t+s)(B + iI)^{-1}\| \leq \|Z(t)(B + iI)^{-1}\|,$$

which proves monotonicity and the upper bound in (5). A known critical-line zero gives an actual eigenvalue with real part $-1/4$, and the fixed-resolvent eigenvector test gives the lower bound. $\square$

Remark 3.2. Norm convergence to zero is strictly stronger than contractive boundedness and is not a consequence of spectral location alone. It supplies no negative exponential exponent: even normal strongly stable semigroups with compact resolvent can have regularized decay only of order $1/t$.

4. EXACT SCALAR REDUCTIONS: TAILS, QUOTIENT NORMS, AND PICK MATRICES

Let Θ be a pure Blaschke inner function on $\mathbb{C}_+$, let

$$A_t = P_{K_\Theta} M_{e^{-ts}}|_{K_\Theta},$$

and let B_Θ generate A_t^* . The results in this section are standard model-space consequences, but their simultaneous application to the regularized zeta model is central to locating the exact obstruction.

Proposition 4.1 (Graph-tail formula). *Let $\mathcal{L} : L^2(0, \infty) \rightarrow H^2(\mathbb{C}_+)$ be the unitary Laplace transform and put $\mathcal{K}_\Theta = \mathcal{L}^{-1}K_\Theta$. Assume $-i \in \rho(B_\Theta)$. Then*

$$\mathcal{L}^{-1}A_t^*\mathcal{L}f(x) = f(x+t), \quad \text{Dom } B_\Theta = \mathcal{K}_\Theta \cap W^{1,2}(0, \infty), \quad B_\Theta f = f',$$

and

$$(6) \quad \|A_t^*(B_\Theta + iI)^{-1}\|^2 = \sup_{0 \neq f \in \mathcal{K}_\Theta \cap W^{1,2}} \frac{\int_t^\infty |f(x)|^2 dx}{\int_0^\infty |f'(x) + if(x)|^2 dx}.$$

Proof. Multiplication by e^{-ts} corresponds to the right shift under the Laplace transform, so the adjoint is left translation. The generator and its domain follow from the standard shift generator. Substituting $h = (B_\Theta + iI)f$ gives (6). $\square$

Assume that Θ continues holomorphically and nonvanishingly through $-i$, and set

$$(7) \quad r(s) = \frac{\Theta(s)/\Theta(-i) - 1}{s + i}, \quad q_t(s) = e^{-ts}r(s).$$

The apparent singularity at $-i$ is removable. Up to a unimodular scalar, r is the boundary model kernel $-k_{-i}^\Theta$.

Proposition 4.2 (H^∞ quotient formula). *With the preceding notation,*

$$(8) \quad \|A_t^*(B_\Theta + iI)^{-1}\| = \|P_{K_\Theta} M_{q_t}|_{K_\Theta}\| = \text{dist}_{H^\infty}(q_t, \Theta H^\infty).$$

Proof. Modulo ΘH^2 , the identity

$$(-s - i)r(s) = 1 - \frac{\Theta(s)}{\Theta(-i)}$$

identifies the compressed multiplier by r with the appropriate adjoint resolvent. Taking adjoints and commuting the analytic compressed multipliers gives the first equality. The second is Sarason's commutant-lifting theorem [11]. $\square$

Let a_j range over the zeros of Θ , with multiplicities m_j , and define

$$\psi_t(s) = -\frac{e^{-ts}}{s + i}.$$

At every zero of Θ , the functions q_t and ψ_t have identical jets through order $m_j - 1$.

Theorem 4.3 (Confluent Pick criterion). *For $C > 0$, the following are equivalent:*

- (i) $\|A_t^*(B_\Theta + iI)^{-1}\| \leq C$;
- (ii) *there exists $F \in H^\infty(\mathbb{C}_+)$ with $\|F\|_\infty \leq C$ and*

$$F^{(p)}(a_j) = \psi_t^{(p)}(a_j), \quad 0 \leq p < m_j;$$

- (iii) *for every finite collection of zero blocks, the confluent Pick matrix*

$$(9) \quad \left[\frac{1}{p!q!} \partial_s^p \partial_{\bar{z}}^q \frac{C^2 - \psi_t(s)\overline{\psi_t(z)}}{s + \bar{z}} \right]_{s=a_j, z=a_k} \Big|_{(j,p),(k,q)}$$

is positive semidefinite.

Proof. Proposition 4.2 and the pure Blaschke factorization show that $F - q_t \in \Theta H^\infty$ exactly when the displayed zero jets agree. The equivalence with (9) is the confluent half-plane Nevanlinna–Pick theorem. Passing from every finite problem to a global interpolant uses a normal-family diagonal argument. See [1] for the complete Pick framework. $\square$

For simple zeros, (9) reduces to

$$(10) \quad \left[\frac{C^2 - w_t(a_j)\overline{w_t(a_k)}}{a_j + \bar{a}_k} \right]_{j,k} \succeq 0, \quad w_t(a) = -\frac{e^{-ta}}{a + i}.$$

The one-point minor already forces

$$(11) \quad C \geq \frac{e^{-t \operatorname{Re} a}}{|a + i|}.$$

Consequently the desired choice $C = C_\varepsilon e^{(-1/4+\varepsilon)t}$ for every $\varepsilon > 0$ forces $\operatorname{Re} a \geq 1/4$ for every zero. In the zeta model the reflected zero symmetry then forces equality. Thus even the diagonal Pick conditions already contain RH; the off-diagonal and confluent conditions contain the additional nonnormal uniformity.

5. WHAT COMPLETE POSITIVITY DOES AND DOES NOT PROVIDE

The Pick matrices admit an exact completely positive formulation. This answers the natural operator-algebra question, but also exposes its circularity at the target scale.

Put

$$D_t = A_t^*(B_\Theta + iI)^{-1}.$$

On a zero kernel k_a ,

$$D_t k_a = \overline{w_t(a)} k_a, \quad w_t(a) = -\frac{e^{-ta}}{a + i},$$

and on derivative kernels it acts by the corresponding lower-triangular jet matrix. Thus Theorem 4.3 is equivalently the assertion that this data operator has norm at most C on every finite jet span, with a bound independent of that span.

Proposition 5.1 (Exact UCP reformulation). *Assume K_Θ is infinite dimensional, as in the zeta model. Let $\tilde{K}(K_\Theta) = \mathcal{K}(K_\Theta) + \mathbb{C}I$ be the unitization of the compact operators. For $C > 0$ define*

$$\Psi_{t,C}(K + \lambda I) = C^{-2}D_t^*KD_t + \lambda I.$$

Then

$$\Psi_{t,C} \text{ is unital completely positive} \iff \|D_t\| \leq C.$$

In particular, existence of these UCP maps for $C = C_\varepsilon e^{(-1/4+\varepsilon)t}$ is exactly the desired regularized estimate.

Proof. If $Q = I - C^{-2}D_t^*D_t \geq 0$ and $\chi(K + \lambda I) = \lambda$, then

$$\Psi_{t,C}(A) = C^{-2}D_t^*AD_t + \chi(A)Q$$

is a sum of completely positive maps and is unital. Conversely, if $P_n \uparrow I$ is a finite-rank approximate identity, positivity gives

$$0 \leq \Psi_{t,C}(I - P_n) = I - C^{-2}D_t^*P_nD_t.$$

Taking strong limits yields $D_t^*D_t \leq C^2I$. □

Stinespring and Arveson extension therefore do not generate the missing estimate. They represent or extend a map after its positivity has been proved. In the present problem, subunitarity of the one-Kraus map $X \mapsto D_t^*XD_t$ is precisely the desired norm inequality. See [2]; the more general operator-argument interpolation theorem of Muhly–Solel has the same logical form [8].

There is nevertheless an unconditional CP factorization, but only at exponent zero. Let r, q_t be as in (7) and $R_0 = \|r\|_\infty$. On all of $\mathbb{C}_+$,

$$(12) \quad \frac{R_0^2 - q_t(s)\overline{q_t(z)}}{s + \bar{z}} = \frac{R_0^2 - r(s)\overline{r(z)}}{s + \bar{z}} + r(s)\overline{r(z)} \int_0^t e^{-u(s+\bar{z})} du.$$

The first summand is a de Branges–Rovnyak kernel and the second is a Gram kernel. Hence every confluent restriction is positive and

$$\|D_t\| \leq \|r\|_\infty.$$

Trying to extract $e^{-t/4}$ termwise fails: on the critical line the remaining phase diagonal is not contractive in the nonorthogonal Cauchy-kernel Gram metric. Complete positivity retains positivity but loses exactly the oscillatory cancellation needed for decay.

6. FIXED FINITE ROOT DIMENSION IS HARMLESS; UNIFORM DIMENSION IS NOT

The following candidate observation isolates the infinite-dimensional nature of the obstruction. It also shows why checking finitely many ordinary Pick matrices, or only close pairs, cannot settle the estimate.

Proposition 6.1 (Uniform estimate at fixed total jet dimension). *Let $V \subset \text{Dom } B$ be a subspace of dimension n for a contraction semigroup, assume $B(V) \subset V$, and suppose*

$$\sigma(B|_V) \subset \{-x + i\eta : \eta \in \mathbb{R}\}$$

for some $x > 0$. For every $\varepsilon > 0$ there is a constant $C_{n,\varepsilon,x}$, independent of spectral heights, gaps, and the Jordan configuration inside V , such that

$$\left\| e^{tB|_V} (B|_V + iI)^{-1} \right\| \leq C_{n,\varepsilon,x} e^{(-x+\varepsilon)t}, \quad t \geq 0.$$

Consequently every critical Pick/Hermite test of a fixed total jet dimension passes with an ε -loss. The constant is not uniform as $n \rightarrow \infty$.

Proof. In an orthonormal Schur basis,

$$B|_V = -xI + iD + U,$$

where D is real diagonal and U is strictly upper triangular. Dissipativity gives $|U_{jk}| \leq 2x$ from each two-dimensional principal minor of $B + B^* \leq 0$, and hence $\|U\| \leq Cxn$. In the interaction picture $U(s) = e^{-isD} U e^{isD}$ remains strictly upper triangular. Every product of n such matrices vanishes, so the Dyson series terminates:

$$\|e^{tB|_V}\| \leq e^{-xt} \sum_{k=0}^{n-1} \frac{(Cxn)^k}{k!}.$$

Put $D_0 = \text{diag}(-x + i(\eta_j + 1))$. Since $B|_V + iI = D_0 + U$ and $D_0^{-1}U$ is strictly upper triangular,

$$(B|_V + iI)^{-1} = \sum_{k=0}^{n-1} (-D_0^{-1}U)^k D_0^{-1}, \quad \|(B|_V + iI)^{-1}\| \leq \frac{1}{x} \sum_{k=0}^{n-1} (Cn)^k.$$

Multiply the two estimates and absorb the fixed polynomial into $e^{\varepsilon t}$. $\square$

For $n = 2$ this yields, uniformly as two critical ordinates coalesce and also for a length-two Jordan block,

$$\|e^{tB|_V} (B|_V + iI)^{-1}\| \leq C e^{-xt} (1 + t).$$

Thus pairwise small gaps cause only polynomial transient amplification. The actual problem is a bound uniform over arbitrarily large and possibly nonlocal zero families, together with unbounded jet orders.

7. WEIGHTED OBSERVABILITY, LYAPUNOV OPERATORS, AND SQUARE FUNCTIONS

Complete positivity becomes useful conceptually when organized as a weighted observability problem. It still gives an equivalent certificate rather than an independent proof.

Let $T(t)$ be a C_0 contraction semigroup with generator B , and let $R \in \mathcal{B}(\mathcal{H})$ commute with $T(t)$. In the application one may take $R = (B + iI)^{-1}$ or any other fixed generator resolvent, since fixed resolvent points give equivalent exponential exponents.

Theorem 7.1 (Weighted Gramian and regularized decay). *Fix $\beta > 0$. The following are equivalent:*

(i) *there is $M_\beta < \infty$ such that*

$$(13) \quad \int_0^\infty e^{2\beta t} \|T(t)Rh\|^2 dt \leq M_\beta \|h\|^2 \quad (h \in \mathcal{H});$$

(ii) *there is a bounded positive operator P_β satisfying, weakly on $\text{Dom } B \times \text{Dom } B$,*

$$(14) \quad \langle P_\beta Bx, y \rangle + \langle P_\beta x, By \rangle + 2\beta \langle P_\beta x, y \rangle = -\langle Rx, Ry \rangle.$$

Either condition implies

$$(15) \quad \|T(t)R\| \leq C_\beta e^{-\beta t}.$$

Conversely, a pointwise estimate with exponent $-\beta'$ for some $\beta' > \beta$ implies (13). Hence the full family (1) is equivalent to existence of such bounded P_β for every $0 < \beta < 1/4$.

Proof. From (13), define the bounded positive Gramian weakly by

$$P_\beta = \int_0^\infty e^{2\beta t} T(t)^* R^* R T(t) dt.$$

Differentiating its shifted weak form at zero gives (14). Conversely, integrate the derivative of $e^{2\beta t} \langle P_\beta T(t)x, T(t)x \rangle$ and use positivity.

Because R commutes with $T(t)$ and $T(t)$ is contractive, the positive operators $(T(t)R)^*T(t)R$ decrease in the Loewner order. Integrating over $[t-1, t]$ gives

$$P_\beta \succeq e^{2\beta t} \left(\int_0^1 e^{-2\beta u} du \right) (T(t)R)^*T(t)R,$$

which proves (15). The converse follows by direct integration with the strict margin $\beta' > \beta$. For the epsilon family, choose $\beta < \beta' < 1/4$ in one direction and $\beta = 1/4 - \varepsilon$ in the other when $0 < \varepsilon < 1/4$. For $\varepsilon \geq 1/4$, contractivity and boundedness of R give the required nondecaying or growing bound directly. $\square$

The equality (14) is only a quadratic-form statement; no claim that $P_\beta \text{Dom } B \subset \text{Dom } B^*$ is made. Similarly, $X \mapsto T(t)^*XT(t)$ is CP and subunitary but need not be norm continuous on $\mathcal{B}(\mathcal{H})$.

The outgoing translation representation gives an especially transparent version. If $\mathcal{T}_+\mathcal{K} \subset L^2(\mathbb{R}_-)$ and $g = \mathcal{T}_+f$, then

$$(16) \quad \|Z(t)f\|^2 = \int_{-\infty}^{-t} |g(r)|^2 dr$$

and Fubini gives

$$(17) \quad \int_0^\infty e^{2\beta t} \|Z(t)f\|^2 dt = \frac{1}{2\beta} \int_{-\infty}^0 (e^{-2\beta r} - 1) |g(r)|^2 dr.$$

Thus the missing estimate is precisely boundedness, after the fixed resolvent, of the exponentially weighted outgoing map

$$h \mapsto e^{-\beta r} \mathcal{T}_+Rh \quad \text{into } L^2(\mathbb{R}_-), \quad \beta < \frac{1}{4}.$$

By vector-valued Paley–Wiener, this is equivalently a vertical-line L^2 resolvent estimate for the regularized product $R(z, B)R$. It is not an absolute L^1 operator-norm contour estimate.

8. FUNCTIONAL-EQUATION SYMMETRY GIVES A KREIN FORM, NOT A POSITIVE METRIC

The zero symmetry $a \mapsto 1/2 - \bar{a}$ suggests shifting the generator by $1/4$ and seeking detailed balance. The idea fails for a structural reason.

Proposition 8.1 (Positive detailed balance annihilates off-critical roots). *Suppose $P \in \mathcal{B}(\mathcal{K})$ is nonnegative and satisfies the weak identity*

$$(18) \quad B^*P + PB = -\frac{1}{2}P,$$

equivalently

$$Z(t)^*PZ(t) = e^{-t/2}P.$$

Then P annihilates every root space with $\text{Re } \lambda \neq -1/4$. On a critical Jordan space it annihilates the nilpotent range. Consequently an injective positive solution would force all zeta roots to be critical and semisimple; in the scalar model it would imply RH together with simplicity of every zero.

Proof. If $Bv = \lambda v$, taking the quadratic form of (18) on v gives

$$\left(2 \text{Re } \lambda + \frac{1}{2} \right) \langle Pv, v \rangle = 0.$$

For an off-critical eigenvalue, positivity gives $Pv = 0$. Induction along a Jordan chain proves the first assertion. On a critical block, the induced nilpotent operator on the quotient by $\ker P$ would be skew-adjoint in a positive metric, hence zero. $\square$

Bare existence of such a P is uninformative: $P = 0$ works, and a known critical adjoint eigenvector produces a nonzero rank-one solution that simply annihilates all other dynamics. A bounded coercive solution would imply the strictly stronger unregularized estimate

$$\|Z(t)\| \leq Ce^{-t/4},$$

which is not needed for the regularized criterion.

On an off-critical reflected eigenpair, every nondegenerate Hermitian solution of the centered invariance equation has, in root coordinates, the form

$$(19) \quad \begin{pmatrix} 0 & c \\ \bar{c} & 0 \end{pmatrix},$$

with signature $(1, 1)$. For paired Jordan blocks the corresponding anti-diagonal form has signature (m, m) . Thus the functional equation naturally supplies a Krein pairing whose cross term decays at the centered rate, not a positive Hilbert metric. Replacing the form by its absolute value or square makes it positive but destroys the shifted intertwining relation.

There is a canonical positive substitute in the time model:

$$Q_\beta(f, g) = \int_0^\infty e^{2\beta x} f(x) \overline{g(x)} dx.$$

On its natural domain,

$$(20) \quad Q_\beta((B + \beta)f, g) + Q_\beta(f, (B + \beta)g) = -f(0)\overline{g(0)}.$$

The negative boundary term is outgoing flux. Graph boundedness of Q_β for every $\beta < 1/4$ is exactly the weighted formulation of (1); the endpoint $\beta = 1/4$ excludes critical eigenfunctions and is strictly too strong. Positivity of the form is automatic, while its required graph boundedness is the unsolved estimate.

9. ABSOLUTE OPERATOR-NORM CONTOUR INTEGRATION IS IMPOSSIBLE

The specification correctly warned that taking the operator norm before a resolvent contour integral loses too much cancellation. In the modular model this warning can be upgraded to a theorem.

Theorem 9.1 (Divergence of the absolute resolvent contour). *Fix $\varepsilon > 0$, put $\alpha = -1/4 + \varepsilon$, and let $\lambda_0 \in \rho(B)$. If the vertical line $\alpha + i\mathbb{R}$ is contained in $\rho(B)$, then*

$$(21) \quad \int_{\mathbb{R}} \|R(\alpha + i\tau, B)^2 R(\lambda_0, B)\| d\tau = \infty.$$

If the line meets the spectrum, the proposed contour is already obstructed by a spectral singularity, which is a pole in the verified meromorphic modular model. Thus no proof of (1) can be obtained by first taking the $\mathcal{B}(\mathcal{K})$ -norm and then estimating this contour absolutely.

Proof. Let $\lambda_n = -1/4 + i\eta_n$ be a simple critical eigenvalue and let v_n be a corresponding unit eigenvector. The verified model correspondence gives such eigenvalues from simple critical zeta zeros. Applying the three resolvents to v_n gives

$$(22) \quad \|R(\alpha + i\tau, B)^2 R(\lambda_0, B)\| \geq \frac{1}{(\varepsilon^2 + (\tau - \eta_n)^2) |\lambda_0 - \lambda_n|}.$$

An unconditional positive proportion of zeta zeros are simple and lie on the critical line; Heath–Brown’s theorem already suffices [4]. Combining this with the Riemann–von Mangoldt formula and the standard $O(\log T)$ bound for the number of zeros in an interval of fixed length, choose one fixed, sufficiently large $q > 1$. In each large geometric block $[T, qT]$ one may select at least cT such ordinates $\eta_n = \gamma_n/2$ with a fixed mutual separation. Trimming the endpoints, and using alternate blocks if necessary, makes all selected intervals disjoint. Integrate (22) over disjoint

fixed-radius intervals centered at the selected ordinates. Since $|\lambda_0 - \lambda_n| \ll T$ in the block, every interval contributes $\gg T^{-1}$. Each geometric block therefore contributes a positive constant, and summing the blocks proves (21). $\square$

The failure is specific to absolute L^1 control of the operator norm. It does not rule out vector-valued L^2 contour or square-function methods. In fact, Section 7 shows that an L^2 estimate with the regularizer retained is exactly the weighted-observability formulation of the desired bound.

10. VERIFICATION OF THE IDENTIFICATION CHAIN

This section records which historical and analytic identifications were actually checked. They are not consequences of formal similarity of scattering coefficients; the precise spaces, factors, signs, and multiplicities matter.

10.1. The reduced zeta factor is unconditionally pure inner. We use the entire completed zeta function

$$\xi(z) = \frac{1}{2} z(z-1) \pi^{-z/2} \Gamma(z/2) \zeta(z)$$

and set

$$(23) \quad \Theta_0(s) = \frac{\xi(2s)}{\xi(1+2s)} = \frac{\xi(2s)}{\xi(-2s)}, \quad \operatorname{Re} s > 0.$$

Theorem 10.1 (Unconditional inner-factor theorem). *The function Θ_0 belongs to $H^\infty(\mathbb{C}_+)$, has norm one, and is a meromorphic inner function of zero mean type with no singular inner factor. Its zeros, with multiplicity, are precisely $a = \rho/2$, where ρ ranges over the nontrivial zeros of ξ . With factors normalized to equal one at the origin,*

$$(24) \quad \Theta_0(s) = \prod_{\rho} \frac{\bar{\rho} \rho - 2s}{\rho 2s + \bar{\rho}}.$$

The product is taken with multiplicity, with the zeros ordered by increasing $|\operatorname{Im} \rho|$ and conjugate-paired; it converges locally uniformly on $\mathbb{C}_+$. Moreover, for the standard Eisenstein coefficient c ,

$$(25) \quad c\left(\frac{1}{2} + s\right) = \frac{s + 1/2}{s - 1/2} \Theta_0(s).$$

No form of RH is used in these assertions.

Proof. The denominator in (23) is nonzero in $\mathbb{C}_+$ because $\operatorname{Re}(1+2s) > 1$. Thus the zero divisor and multiplicities follow without cancellation. Lagarias's modified-Hadamard-product inequality [5, Lemma 2.1] states, for $h \geq 1/2$ and $\operatorname{Re} u > 1/2$, that

$$|\xi(h+u)| > |\xi(h+1-\bar{u})|.$$

Taking $h = 1/2$ and $u = 1/2 + 2s$ gives

$$|\xi(1+2s)| > |\xi(1-2\bar{s})| = |\xi(2s)|,$$

so $|\Theta_0(s)| < 1$ in $\mathbb{C}_+$. On $s = iy$, the functional equation and reality symmetry give $\xi(1+2iy) = \overline{\xi(2iy)}$, with neither side zero. Hence the boundary modulus is one at every finite boundary point.

Lagarias is used here only for the Hermite–Biehler inequality and hence the inner bound. The remaining factorization assertions follow from the boundary-meromorphy and asymptotic argument that follows. The zero counting formula implies the half-plane Blaschke condition. Since Θ_0 is meromorphic across every finite point of $i\mathbb{R}$, the finite boundary singular measure vanishes. Direct algebra and Stirling's formula give, as $x \rightarrow +\infty$,

$$(26) \quad \Theta_0(x) = \sqrt{\pi} x^{-1/2} (1 + O(x^{-1})), \quad \lim_{x \rightarrow \infty} \frac{1}{x} \log |\Theta_0(x)| = 0.$$

Thus the exponential factor at infinity is absent. This proves the pure Blaschke factorization and (24). Finally, substitution of the definitions gives

$$\Theta_0(s) = \frac{s - 1/2}{s + 1/2} \sqrt{\pi} \frac{\Gamma(s)\zeta(2s)}{\Gamma(s + 1/2)\zeta(1 + 2s)},$$

which is (25). $\square$

Here zero mean type means the second limit in (26). Some conventions include the exponential mass at infinity among singular inner factors; the theorem excludes it explicitly.

10.2. Positive energy and the double-primed Lax–Phillips system. Let $X = \mathrm{PSL}_2(\mathbb{Z}) \backslash \mathbb{H}$, use the positive Laplacian

$$\Delta_X = -y^2(\partial_x^2 + \partial_y^2),$$

and set $A_c = (\Delta_X - 1/4I)|_{L_c^2(X)}$. If $\mathcal{E}$ denotes the Eisenstein transform, normalized so that

$$\mathcal{E}A_c\mathcal{E}^{-1} = M_{\kappa^2},$$

then on a smooth absolutely continuous core

$$(27) \quad (f_0, f_1) \longmapsto (\kappa\mathcal{E}f_0, \mathcal{E}f_1)$$

is isometric for the positive wave energy. Completion gives the homogeneous form space

$$\dot{H}_{A_c}^1 \oplus L_c^2(X).$$

The Pavlov–Faddeev absolutely continuous first-order data space and Uetake’s continuous positive-energy space are canonically unitarily equivalent through this common Eisenstein spectral representation; see [10, pp. 164–170] and [13, pp. 102–105]. Thus (27) supplies the required energy-unitary identification. Pavlov–Faddeev use an overall factor 1/2 in the quadratic energy, removed by a harmless scalar normalization.

For the modular surface, the constant mode is the sole below-threshold residual mode. Cusp-form standing waves are removed on passing to the continuous first-order spectral space, and there is no L^2 threshold eigenfunction at 1/4. The absence of a threshold eigenvalue is an explicit assumption in the general Pavlov–Faddeev construction and must not be silently generalized beyond the modular case.

The correct incoming and outgoing spaces for the orthogonal interaction semigroup are

$$\mathcal{D}_{\pm}'' = \mathcal{D}_{\pm}^0 \cap \mathcal{H}'_E,$$

not the projected primed spaces. Uetake verifies invariance, trivial intersection, translate density, and $\mathcal{D}_{-}'' \perp_E \mathcal{D}_{+}''$; see [13, pp. 104–105]. If

$$\mathcal{K} = \mathcal{H}'_c \ominus_E (\mathcal{D}_{-}'' \oplus \mathcal{D}_{+}''),$$

then, for $t \geq 0$,

$$(28) \quad Z(t) = P_{\mathcal{K}}^E U(t)|_{\mathcal{K}} = Q_{+} U(t) Q_{-}|_{\mathcal{K}}$$

is a strongly continuous contraction semigroup. Orthogonality and the one-sided invariance give (28) directly; the discarded outgoing component remains outgoing, which proves the semigroup law. Pavlov–Faddeev prove the analogous orthogonal Lax–Phillips axioms and contraction semigroup for their own incoming and outgoing spaces [10, pp. 163–175]; see also Lax–Phillips [7, §5].

10.3. The full factor, reduced model, and exceptional mode. Put

$$b_{1/2}(s) = \frac{s - 1/2}{s + 1/2}, \quad \Phi(s) = -b_{1/2}(s)\Theta_0(s).$$

For the orthogonal double-primed system, the full scattering factor is

$$(29) \quad S_c''(s) = \Phi(s) = -\frac{s - 1/2}{s + 1/2} \frac{\xi(2s)}{\xi(-2s)},$$

not $c(1/2 + s)$ and not Θ_0 alone [13, p. 111]. The reciprocal finite factor appearing for Uetake's nonorthogonal primed spaces is a different convention [13, pp. 117–118]. In this subsection write $\mathcal{K}'' = \mathcal{K}$ for the full physical double-primed interaction space.

Theorem 10.2 (Exact model and finite exceptional splitting). *Let $\mathcal{T}_+$ be Uetake's outgoing energy-unitary translation map and normalize*

$$[\mathcal{L}_-g](s) = \int_{-\infty}^0 e^{s\tau} g(\tau) d\tau, \quad \frac{1}{2\pi} \int_{\mathbb{R}} |\mathcal{L}_-g(iy)|^2 dy = \|g\|_2^2.$$

Then

$$(30) \quad [\mathcal{U}_{\text{full}}f](s) = [\mathcal{L}_-\mathcal{T}_+f](s) = \int_0^\infty e^{-sx} (\mathcal{T}_+f)(-x) dx, \quad \mathcal{U}_{\text{full}} : \mathcal{K}'' \longrightarrow K_\Phi.$$

is unitary and intertwines $Z(t)$ with

$$A_{\Phi,t}^* = (P_{K_\Phi} M_{e^{-ts}}|_{K_\Phi})^*.$$

There is a bounded invariant topological direct sum

$$(31) \quad K_\Phi = K_{\Theta_0} \dot{+} \text{span}\{k_{1/2}^\Phi\},$$

under which

$$(32) \quad Z(t) \text{ is boundedly similar to } A_{0,t}^* \oplus e^{-t/2}.$$

More explicitly, if

$$J : K_{\Theta_0} \oplus \mathbb{C} \longrightarrow K_\Phi, \quad J(f, c) = f + c k_{1/2}^\Phi,$$

then the exact physical-to-split-model intertwiner is

$$(33) \quad W = J^{-1} \mathcal{U}_{\text{full}}.$$

The reduced generator has spectrum, with algebraic multiplicity,

$$(34) \quad \sigma(B_0) = \{-\bar{\rho}/2 : \xi(\rho) = 0\} = \{-\rho/2 : \xi(\rho) = 0\},$$

its generalized root vectors are complete, and the full generator adds only the one-dimensional eigenvalue $-1/2$. There are no other finite spectral points, and $-i \in \rho(B)$.

Proof. The unitary full-model transform and the generator convention are given in [13, pp. 109–113]. Because $b_{1/2}\Theta_0$ has one more Blaschke zero than Θ_0 , K_{Θ_0} has codimension one in K_Φ and is invariant under the adjoint model semigroup. The reproducing kernel at $1/2$ is an eigenvector with orbit $e^{-t/2}k_{1/2}^\Phi$ and is not in K_{Θ_0} , since $\Theta_0(1/2) \neq 0$. It therefore supplies the invariant complement in (31). The splitting is bounded because it is a closed codimension-one decomposition, but is not in general orthogonal. Thus J is boundedly invertible, and (33) proves (32).

Uetake's Theorems 4.2 and 4.4 identify the reduced resolvent poles, algebraic multiplicities, and complete root system. The two forms of the set in (34) agree by the functional-equation and conjugation symmetries. The separate full-space statement [13, pp. 100–101] gives $\sigma(B'') \setminus \sigma(B_0) = \{-1/2\}$ and says that the full space is one dimension larger. All nontrivial zeta zeros lie in $0 < \text{Re } \rho < 1$, so their generator eigenvalues have real parts strictly between $-1/2$ and 0 ; the exceptional eigenvalue is $-1/2$. These results exclude all other finite spectrum and prove $-i \in \rho(B)$. $\square$

The first transform in Theorem 10.2 is unitary. The second, finite-dimensional splitting is only a bounded similarity. Hence it gives two-sided norm comparison and exact equality of logarithmic exponents, not literal equality of finite-time norms. The exceptional block decays like $e^{-t/2}$ and cannot change an exponent at or above $-1/4$.

10.4. The primary Pavlov–Faddeev theorem. The primary 1972 paper, in its English translation [10, §§3–4], does prove a whole-space, resolvent-regularized strip theorem. Theorem 1 relates bounded holomorphic continuation of the rationally regularized scattering function to exponential decay of the uniform interaction-space operator norm [10, original pp. 177–180]; the converse holds in each strictly smaller strip. Lemma 8 [10, pp. 185–186] supplies the zeta estimates in every strip of width $a < 1/4$ under RH, and the corollary, formulas (48)–(49) [10, pp. 186–187], states the logarithmic limsup $-1/4$ with the quantifiers

$$\forall a < \frac{1}{4} \exists C_a \forall t \geq 0.$$

The paper therefore supports exactly the regularized operator criterion, not an unregularized norm or a dense fixed-vector substitute.

Its native convention is $Z(t) = e^{itB_{\text{PF}}}$. If $Z(t) = e^{tB}$ in the ordinary generator convention, then $B = iB_{\text{PF}}$ and

$$(35) \quad (B_{\text{PF}} + iI)^{-1} = -i(I - B)^{-1} = -iR(1, B).$$

Only after verifying $1, -i \in \rho(B)$ and applying the fixed-resolvent comparison may this be replaced, at the exponent level, by $(B + iI)^{-1} = -R(-i, B)$. Thus the logarithmic exponent and the epsilon family are unchanged, although the displayed resolvents are not literally equal. The secondary survey [12, §3.2, (3.6), p. 1004] quotes the same regularized formula; the native convention is checked in the primary paper above.

11. OTHER PARTIAL RESULTS AND ROUTES THAT WERE TESTED

11.1. The diagonal zero test has stretched-exponential decay. The fixed regularizer suppresses very high zeros strongly. The classical zero-free region, combined with the functional equation, gives

$$\operatorname{Re} a \geq \frac{c}{\log(2 + |\operatorname{Im} a|)} \quad (a = \rho/2)$$

outside a bounded set. Optimizing in height yields the unconditional one-point estimate

$$(36) \quad \sup_{\Theta_0(a)=0} \frac{e^{-t \operatorname{Re} a}}{|a + i|} \leq C e^{-c' \sqrt{t}}.$$

This is a useful explanation of the regularizer, but it controls only the 1×1 Pick minors. It gives no upper bound for the full operator norm, which depends on arbitrary linear combinations and confluent jets.

11.2. Ordinary Riesz-basis and Carleson synthesis are unavailable. On the critical line, the normalized kernel inner product for two nodes $a = 1/4 + i\gamma/2$ and $b = 1/4 + i\gamma'/2$ has modulus

$$\frac{1}{\sqrt{1 + (\gamma - \gamma')^2}}.$$

The Riemann–von Mangoldt density together with the unconditional positive proportion of simple critical zeros forces arbitrarily close distinct nodes, so the normalized root kernels are not a Riesz sequence. More explicitly, the standard atomic measure

$$\mu = \sum_j 2 \operatorname{Re}(a_j) \delta_{a_j}$$

fails the half-plane Carleson condition: unit boundary intervals along a sequence contain $\gg \log T$ simple critical nodes. Resolvent weighting may restore an upper Bessel–Carleson bound, but it supplies neither a lower Riesz bound nor a bounded coefficient-extraction map. It therefore does not identify the graph norm with a weighted ℓ^2 coefficient norm.

11.3. Compactness and strong stability have no quantitative rate. The proof of Theorem 3.1 uses all the available qualitative model information. It cannot be sharpened abstractly. For example, on $\ell^2(\mathbb{N})$ let

$$Be_n = (-1/n + in)e_n.$$

Then e^{tB} is a normal, strongly stable contraction semigroup, its eigenvectors are an orthonormal basis, and $R(1, B)$ is compact, but

$$\|e^{tB}R(1, B)\| \asymp t^{-1}.$$

Thus even normality, complete roots, compact regularization, and strong stability give exponent zero without additional quantitative geometry.

11.4. Why functional-equation pairing cannot be positivized. There are two distinct reflections. Scattering gives

$$\Theta_0(-s) = \Theta_0(s)^{-1},$$

which is centered on the Hardy boundary $\operatorname{Re} s = 0$. Zeta-zero symmetry gives $a \mapsto 1/2 - \bar{a}$, centered on $\operatorname{Re} a = 1/4$. The latter does not preserve $\mathbb{C}_+$ or its boundary and does not define a bounded Hardy-space conjugation. The canonical model conjugation yields $B \leftrightarrow B^*$ but no $1/4$ shift. The algebraic critical reflection yields the indefinite form (19); applying Q^*Q or $|Q|$ removes exactly the off-diagonal relation that encoded the functional equation.

12. CORRECTIONS AND QUALIFICATIONS TO THE GOVERNING SPECIFICATION

No error was found in the graph-norm equivalence, fixed-resolvent comparison, formula (25), adjoint/time sign, eigenvector lower bound, or the epsilon-family quantifiers. The following points should nevertheless be corrected or sharpened.

- (S1) The five labeled identification hypotheses can now be replaced by explicit propositions and primary citations, as summarized in Section 10. They must not remain implicit in a claimed proof.
- (S2) The physical double-primed scattering factor is $-b_{1/2}\Theta_0$. The coefficient $c(1/2 + s) = b_{1/2}^{-1}\Theta_0$ has a pole at $s = 1/2$, whereas the orthogonal double-primed factor has a zero there and produces the generator eigenvalue $-1/2$. These are different objects.
- (S3) The hypothesis that Θ_0 is bounded inner, of zero mean type, and without a singular factor is an unconditional theorem (Theorem 10.1), not an RH-dependent assumption.
- (S4) In the general Pavlov–Faddeev construction absence of a threshold eigenvalue is an explicit assumption. It is valid for the modular surface after the threshold analysis, but not automatically for an arbitrary one-cusp group.
- (S5) Pavlov–Faddeev’s energy differs by an overall factor $1/2$ from the modern convention. A scalar rescaling makes the identification literally unitary; this does not affect operator norms after both source and target are rescaled consistently.
- (S6) A general bounded similarity provides two-sided constant comparison of norms. Unitarity guarantees exact equality, but it is too strong to say that exact equality can occur only under unitarity.
- (S7) The exceptional spectral-bound argument uses finite dimensionality: on the one-dimensional exceptional block the growth exponent equals the spectral bound. Such an inference is false for a general infinite-dimensional semigroup.

- (S8) The phrase zero mean type should be defined, for example by (26). Boundary modulus alone never proves an H^∞ bound.
- (S9) The specification’s heuristic warning about absolute contour integration can be replaced, under the verified model, by Theorem 9.1.
- (S10) Any assertion that the unregularized model norm is exactly one for all $t > 0$ needs a separate model-space theorem or proof. It is irrelevant to (1) and should not be used as a substitute for the regularized estimate.

13. THE EXACT REMAINING OBSTRUCTION AND ACCEPTANCE AUDIT

For the zeta inner function, put $a_j = \rho_j/2$, retain the multiplicity m_j , and set $\psi_t(s) = -e^{-ts}/(s+i)$. After all physical-to-model identifications and the exceptional splitting, the unresolved assertion is the following single uniform statement:

For every $\varepsilon > 0$ there is one constant C_ε , independent of t , the number and heights of the selected zeros, their gaps, and their multiplicities, such that every finite confluent matrix

$$(37) \quad \left[\frac{1}{p!q!} \partial_s^p \partial_{\bar{z}}^q \frac{C_\varepsilon^2 e^{(-1/2+2\varepsilon)t} - \psi_t(s) \overline{\psi_t(z)}}{s + \bar{z}} \right]_{s=a_j, z=a_k} \Big|_{(j,p),(k,q)} \succeq 0$$

for all $t \geq 0$ and $0 \leq p < m_j$, $0 \leq q < m_k$.

Equivalently, one must prove the graph-tail inequality

$$(38) \quad \int_t^\infty |f(x)|^2 dx \leq C_\varepsilon^2 e^{(-1/2+2\varepsilon)t} \int_0^\infty |f'(x) + if(x)|^2 dx$$

for every $f \in \mathcal{K}_{\Theta_0} \cap W^{1,2}$, with one constant for the whole graph-norm unit ball. It is also equivalent to graph boundedness of all subcritical outgoing weights, or to the bounded positive Lyapunov operators of Theorem 7.1.

The one-point part of (37) is already RH-equivalent after zero symmetry. Assuming RH removes that diagonal obstruction, but it does not prove the full matrices: Theorem 2.1 shows that large confluent blocks can destroy the exponent even when every zero is critical. Proposition 6.1 shows the converse local fact: every fixed total jet dimension is harmless after an ε -loss. The unresolved content is precisely uniformity as the total dimension tends to infinity.

Table 1 records the audit against the ten acceptance criteria of the governing specification. A check means that the issue has been handled in the reductions above; it does not mean that the final uniform estimate has been proved.

Item	Status	Audit
A1	Exact physical energy space and physical-to-model transform are stated in Section 10.	check
A2	The positive reduced energy norm is used; the raw indefinite form is not used.	check
A3	The exact fixed-resolvent graph norm is retained in (6); fixed-resolvent changes are proved harmless.	check
A4	The order $\forall \varepsilon \exists C_\varepsilon \forall t$ appears explicitly in (1) and (37).	open
A5	The Pick, tail, and Lyapunov statements all range over the whole graph-norm unit ball, not a fixed or dense set of vectors.	open
A6	The full factor $-b_{1/2}\Theta_0$ is split into the reduced model plus the faster one-dimensional block.	check
A7	Ordinary generator, time direction, adjoint model, zero map, and bounded norm transfer are tracked in Theorem 10.2.	check
A8	Normal-family and weak Gramian limits are justified for the stated reductions; absolute contour interchange is rejected by Theorem 9.1.	qualified
A9	All uniform root-based criteria retain multiplicities through confluent jets, and the counterexample treats full Jordan blocks.	check
A10	No growth estimate is inferred from spectrum alone; the abstract counterexample proves why this would be invalid.	check

TABLE 1. Acceptance audit. A4–A5 remain open because they are the target uniform estimate itself.

14. CONCLUSION

The historical identifications and the analytic inner-function hypothesis can be discharged rigorously. The exact problem that remains is not a spectral-bound calculation: it is the uniform confluent Pick inequality (37), equivalently the whole-space graph-tail estimate (38) or a bounded subcritical observability Gramian. Standard complete positivity, Stinespring dilation, Arveson extension, scattering unitarity, root completeness, compact resolvent, and the functional equation each illuminate this statement but do not supply its missing uniform constant. The functional equation gives an indefinite critical pairing; positive forms become useful only after imposing the graph boundedness that is itself the target.

The strongest unconditional operator conclusion currently obtained is

$$\|Z(t)(B + iI)^{-1}\| \rightarrow 0, \quad -\frac{1}{4} \leq \omega_{\text{reg}}(B) \leq 0.$$

The critical-line Blaschke construction proves that further abstract model theory alone cannot close this bracket. Any successful proof must introduce new zeta-specific quantitative control of arbitrarily large zero families and their jets, or an equivalent automorphic observability estimate. Conversely, an unconditional proof of that uniform estimate would, already by its one-dimensional minors and zero symmetry, prove RH. No such proof or disproof has been found here.

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